

From CNNs to Transformers: A Comparative Study of Sentiment Analysis on SST-2

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1. Introduction

Sentiment Analysis is a significant activity under Natural Language Processing (NLP). It is a process of categorising the opinions expressed in texts as positive, negative, or neutral. This ability helps determine the extent to which NLP models understand meaning. It has numerous applications in practice: the analysis of customer feedback and monitoring brand reputation. One of the frequently used sources to do this classification task is the Stanford Sentiment Treebank (SST-2) dataset. Researchers use this dataset to train and evaluate models for sentiment analysis.

Approaches to sentiment analysis have evolved over the last years. Previously, deep learning models, particularly Convolutional Neural Networks (CNNs), had achieved moderate success in learning hierarchical features directly from text. The birth of Transformer-based models, particularly BERT (Bidirectional Encoder Representations from Transformers), has been a significant step forward in the scene, however. These models are pre-trained on a large scale using massive and unlabelled text data, which is then fine-tuned to specific tasks. In this way, they can gain a better understanding of the language's context.

This work attempts to explore the performance of various modeling strategies on three levels: (1) a simple deep learning baseline based on a CNN architecture; (2) a pre-trained bert-base-uncased model, which is tested without any form of fine-tuning to test its zero-shot performance; (3) the same transformer, which is modified via a reasonable strategy, a fine-tuning method that reaches out to the layers to balance knowledge retention and task adaptation gradually.

The purpose of this work is to estimate the value of both advanced model architecture and domain-specific adaptation, which are often considered. This research aims to demonstrate the considerable variations in performance between the standard baseline and the pre-trained and fine-tuned transformer and underline the crucial contribution of the latter to achieving the full potential of large language models in specialised applications in modern NLP systems.

2. Related Work

The history of research in sentiment analysis is long, and it has undergone methodological changes along with the rest of Natural Language Processing. In itself, the release of the Stanford Sentiment Treebank (SST-2) was a significant breakthrough, offering a dataset that no longer focused solely on document-level labels but also takes into account the compositional influence of syntax on sentiment [Socher et al., 2013]. Early deep learning systems on this data were primarily based on either Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs). These models facilitated the learning of sentence representations using word embeddings. This particular ability of CNNs to learn valuable local n-gram features that capture the meaning of the sentiment is powerful [Kim, 2014]. These architectures provided very robust baselines; however, in most cases, they were constrained by their reliance on learning with the labelled dataset only.

Large-scale pre-trained models represented a paradigm shift in NLP, and sentiment analysis was no exception. A further breakthrough was made. With the introduction of BERT (Bidirectional Encoder Representations from Transformers), a deep bidirectional model pre-trained on a large unlabelled corpus of text [Devlin et al., 2019]. With the aid of a downstream task such as SST-2, researchers could get new state-of-the-art results with relative ease by merely fine-tuning this pre-trained model. This demonstrated that the contextual word representation learned during the rich pre-training was highly effective in sentiment classification. The pre-training process has also been improved by future advances, including RoBERTa, with even better results on benchmark data [Liu et al., 2019].

Research was also interested in finding more advanced forms of adaptation in order to optimise performance and training stability when fine-tuning became the norm. Standard fine-tuning optimises all the parameters in a model, and can occasionally cause problems such as catastrophic forgetting, in which the model forgets some of its strong, general knowledge learned during pre-training. In response to this, a series of methods, such as progressive unfreezing, were developed. Here, the initial layers of the model are not immediately fine-tuned, but only the last few classification layers, followed by a series of training steps where more and more of the model is unfrozen and refrozen. This enables a more balanced and focused adaptation to the new task, and is an efficient transfer learning strategy in NLP [Howard and Ruder, 2018]. This principle of methodical layer unfreezing is the basis of the third model in this report.

3. Method

This section explains the models and processes involved in performing sentiment analysis. Three methods have been outlined: (1) a classic deep learning baseline, which is a Convolutional Neural Network (CNN), (2) an off-the-shelf, pre-trained transformer, bert-base-uncased, which was not finetuned, and (3) a slightly adapted version of the same transformer, adapted to the task using a specialized finetuning strategy. Every model has subsections that describe its architecture, data pre-processing, and training/evaluation methodology.

3.1 Traditional Deep Learning Baseline CNN.

A Convolutional Neural Network (CNN) was utilized to perform binary classification on the SST-2 dataset, leveraging TensorFlow and Keras to achieve a comparable level of performance.

Pre-processing and Data Tokenisation.

The Keras Documentation assisted in preparing the uncoded data for the SST-2 set. It was created using the 5,000 most frequent words to establish a vocabulary. Converting individual sentences into streams of token IDs.. The sentences were each converted into a list of integer tokens. Each sequence was cut or padded to 50 tokens (the most extended possible duration) so that all sequences had equal amounts of available input to the model.

Model Architecture

The CNN was constructed in the form of a Friday-ladder, and the layers were:

- An Embedding layer that maps integer token IDs into a 50-dimensional embedding space of vectors.
- A 1D Convolutional layer with a fifth-size kernel and 64 filters alongside a \$ReLU activation function so that local features in the text are detected.
- GlobalMaxPooling1D layer to sample the feature maps and only extract salient features.
- Dense: An output layer that has one unit and applies the sigmoid activation function to give a probability measure of the positive class.

Training and Evaluation

The binary cross-entropy loss function is implemented in conjunction with the Adam optimizer. These techniques were typically used in binary classification and were used to compile the model. The batch size was 64, and the model was trained for three epochs. The validation set was used to evaluate performance, with the key metric being accuracy.

3.2 Off-the-Shelf Transformer: BERT-base- uncased.

The SST-2 test was used to conduct the baseline test of the BERT-base-uncased model without any finetuning. This experiment aimed to measure the extent to which this model possesses the property of understanding sentiment solely through its large pre-training.

Pre-processing, Model and Tokenization.

The BERT-base-uncased model and tokenizer are loaded directly into the Hugging Face Transformers library. Sentences were tokenized, truncated or padded to 128 tokens (SST-2). The converted information, input_ids, attention_mask, and labels, was once again converted into PyTorch Tensors for evaluation. This was not divided into training.. There was no training split on this step.

Evaluation Setup

The pre-trained model was evaluated using the Hugging Face Trainer API, with training explicitly disabled (do_train=False). An evaluation batch had a size of 8. Performance evaluation was conducted in terms of accuracy and weighted F1-score, calculated via a custom compute_metrics function.

3.3 Fine-Tuned Transformer with Advanced Strategies.

The last methodology involved finetuning the pre-trained BERT-base-uncased model on the SST-2 data to achieve the best results without overfitting.

Dataset and Pre-processing

The tokenization process and model used in this step were identical to those described in Section 3.2. In this stage, however, the training set was

used. To ensure computational efficiency, a random sample of 5,000 samples from the official SST-2 training set was finetuned.

Overfitting minimization and finetuning Strategy.

Among the major decisions made during the training setup, the idea of conducting only a single whole epoch of the finetuning process is worth mentioning. As complex as the BERT architecture, the model is likely to overfit when trained on a comparatively small, task-specific problem, such as SST-2. The extended training will force the model to memorize the specifics of the training data, beyond generalizable trends in sentiment. Training one period of time was, therefore, a deliberate decision to provide sufficient time to the model to adapt to the task and eliminate the risk of overfitting.

This core strategy was supported by many complementary techniques such as:

- **Progressive freezing:** To freeze the lower six layers of the BERT encoder in the first (and only) training pass, a custom `TrainerCallback` was used that applied a layer-at-a-time strategy, where the weights of empty layers were fixed and all upper, more task-specific layers were allowed to train.
- **Regularisation:** To further deter high-valued weights and minimize the disposition towards overfitting, we used the optimizer with regularisation (weight decay = 0.01).
- **Early Stopping:** To validate the metric, `EasyStoppingCallback` was added to control when the training process can be stopped, and the time when this is done is the most appropriate, thus avoiding unnecessary calculations.

Training Configuration

The AdamW optimizer was used to develop the model with a learning rate of 2×10^{-5} . The gradient accumulation step sizes of 2 and 4 per device were found to be sufficient to achieve a batch size of 8. Mixed-precision training (FP16) was also made available to achieve higher memory efficiency and speed. The final model checkpoint was selected based on the maximum accuracy achieved during the validation process of the training activity.

4. Result

The goal of the experiments was to provide a clear comparison of how three different approaches to the SST-2 sentiment analysis task stack up against a traditional CNN baseline. We looked at an off-the-shelf Transformer model, as well as a fine-tuned Transformer model, to see how they performed in this context.

Model / Approach	Val Loss	Val Accuracy (%)	Weighted F1
CNN Baseline	0.2807	80.16	0.8161
BERT-base (Off-the-shelf)	0.7470	49.08	0.3232
BERT-base (Fine-tuned)	0.2882	90.14	0.9014

Table 1: The final validation of all the models with the SST-2 data. The CNN is taken as a classical point of reference, and the BERT model is off-the-shelf and fine-tuned.

4.1 CNN Baseline Performance

The CNN trained on the entire SST-2 training set reached the final accuracy of 80.16% on the validation set. This finding represents an impressive performance of a standard neural framework, which means that the model learns beneficial sentiment features, i.e., keywords and n-grams, via supervised billing. But this method is restricted, naturally, to trends discussed within training data and does not examine the more contextual view of contemporary architectures. However, this approach is confined, of necessity, to the patterns deliberated in the training data and does not explore the more contextual knowledge of modern architectures.

4.2 Off-the-Shelf Transformer Performance

The off-the-shelf BERT-base-uncased model was tested without fine-tuning and achieved an accuracy of only 49.08%. This is no better than the performance of a binary classification task of random chance. This is unsurprising and suggests a critical point. However, the pre-trained Transformer possesses an enormous amount of general language knowledge; its classification head is randomly chosen and has no prior training to perform a specific task. The model cannot learn its representations to the exact labels of sentiment needed by SST-2 without fine-tuning.

4.3 Fine-Tuned Transformer Performance

With a single epoch of fine-tuning on a specialised strategy, the BERT-base-uncased model's performance has significantly increased, reaching an ultimate validation accuracy of 90.14%. This huge growth of more than 40 percentage points above the state-of-the-art highlights the effectiveness of transfer learning to the utmost. The findings validate that new state-of-the-art levels of performance can be achieved through the training of pre-trained weights for the particular sentiment analysis task, which utilises their rich, contextual representations.

4.4 Comparative Analysis

The results provide some important lessons about the different approaches to sentiment analysis:

- The CNN model itself is a respectable baseline, showing that in the supervised scenario, traditional deep learning models can be learned successfully; however, their performance cannot be further enhanced due to the limitations of the architecture.
- This classification cannot be directly done using an off-the-shelf Transformer. Its performance has not been good, which demonstrates that the pre-training is not sufficient and that the final layers of the network must be adapted to the task at hand.
- The fact that it is fine-tuning is the most crucial aspect, as it unlocks the capabilities of pre-trained models. The radical increase in performance shows that transferring general knowledge of a Transformer to a specific domain, when trained on a minimal number of examples, is a highly effective and superior approach to NLP tasks.

5. Discussion

The purpose of the project was to demonstrate the practical importance of various modelling paradigms, i.e., a comparison between the traditional approach of deep learning and fine-tuning transformers, in a typical sentiment analysis task. These findings yield conclusive results, showing that adapting tasks through fine-tuning yields a greater output in terms of enhancing performance compared to conventional architectures and commercially available transformers.

It was also informative in terms of the performance difference between the two uncased models of Bert-base. The ineffectiveness of the off-the-shelf variant was noticeable, as the accuracy was only 49.08, which represented the results of random choice. This implies that the final classification layer is not created when training an example; a learned and powerful model can make no use of it directly in presenting a specific classification problem. Nevertheless, after subjecting a small sample of SST-2 to further refinements with the same model, the model accuracy increased to 90.14. This supports the claims that optimisation of tasks applies even to strong pre-trained representations.

The transformer, which is one of the classical approaches to deep learning, ultimately defeated the CNN model, whose initial accuracy was 80.16. This brings a natural susceptibility to the transformer architecture to NLP. The self-attention mechanism can learn longer-range dependencies and other intricate contextual relations in language compared to a CNN that learns local features.

Another interesting aspect of the findings was the correlation between loss of validation and accuracy. Although the fine-tuned BERT had the best accuracy (90.14%), it had a lower validation loss (0.2882) when compared to the less accurate CNN baseline (0.2807). It implies that, although the BERT model is less error-prone, it tends to be even more confident in making its errors, which the cross-entropy loss function pays for with even more scores. It means that the fine-tuned model may be slightly less calibrated, but the fact that it exhibits a higher correctness rate is what makes it a much better model overall.

6. Conclusion

The project performed a comparative study on two additional progressive levels of the SST-2 data sentiment model. As illustrated, the experiments demonstrated a clear order of magnitude in performance, with a working CNN baseline outperforming an off-the-shelf transformer and a high-performing, fine-tuned transformer. Its findings demonstrate that task adaptation is a key factor that can be used to utilise huge pre-trained language models.

Fine-tuning has been significantly optimised, which confirms that transfer learning is a very efficient and effective solution in contemporary NLP. The observations may be of assistance in the work to come because they emphasise the fact that, despite very specialised architectures, there is always a need to fine-tune them. This study can be extended in a future study with a more detailed examination of larger models, such as RoBERTa, or by investigating the effects of various hyperparameters and fine-tuning strategies on the functionality and calibration of the final model.

7. References

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